

Distributed Non-Uniform Scaling Control of Multi-Agent Formation via Matrix-Valued Constraints

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Abstract—Distributed formation maneuver control refers to the problem of maneuvering a group of agents to change their formation shape by adjusting the motions of partial agents, where the controller of each agent only requires local information measured from its neighbors. Although this problem has been extensively investigated, existing approaches are mostly limited to uniform scaling transformations. This article proposes a new type of local matrix-valued constraints, via which non-uniform scaling control of position formation can be achieved by tuning the positions of only two agents (i.e., leaders). Here, the non-uniform scaling transformation refers to scaling the position formation with different ratios along different orthogonal coordinate directions. Moreover, by defining scaling and translation of attitude formation, we propose a distributed control scheme for scaling and translation maneuver control of joint position-attitude formations. It is proven that the proposed controller achieves global convergence, provided that the sensing graph among agents is a 2-rooted bidirectional graph. Compared with the affine formation maneuver control approach, the proposed approach leverages a sparser sensing graph, requires fewer leaders, and additionally enables scaling transformations of the attitude formation. A simulation example is proposed to demonstrate our theoretical results.

Index Terms—Non-uniform scaling, matrix-valued constraint, 2-rooted graph, distributed formation control, multi-agent systems.

I. INTRODUCTION

Formation maneuver control enables a group of agents to operate as a cohesive unit, with maneuverability defined as the degree to which the formation's positional (centroid, scale, and other geometric parameters) and attitudinal (orientation) characteristics can be continuously adjusted while maintaining coordinated motion. This capability is essential for applications such as search and rescue [1], [2], cooperative transport [3], cooperative localization [4], and collaborative manipulation [5]. However, dynamic and complex environments, such as obstacle-dense or high-interference scenarios, pose significant challenges to formation maneuverability.

The maneuverability of a multi-agent position formation is fundamentally constrained by the types of local inter-agent constraints that characterize the overall formation geometry. Prior studies have demonstrated translational maneuvers via displacement-based consensus methods [6], [7], and rotational maneuvers through inter-agent distance constraints [8]–[10]. Compared to rigid transformations only, scalable formations

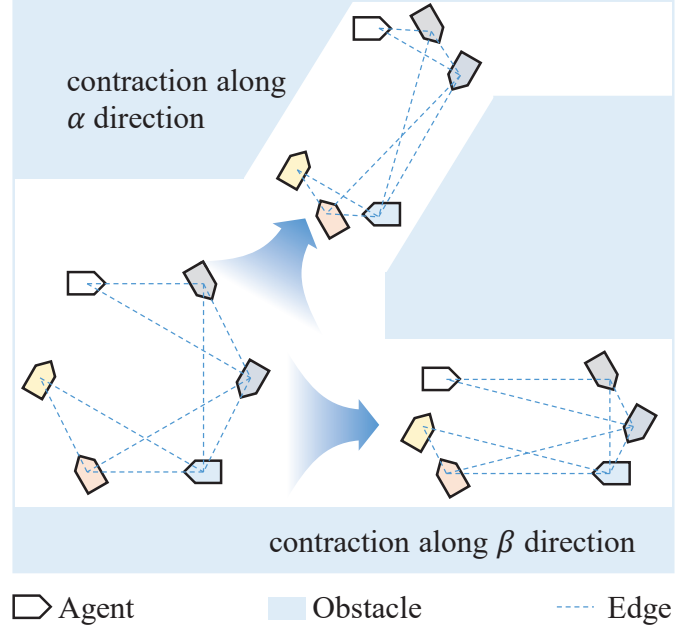


Fig. 1: Non-uniform scaling transformation along arbitrary direction of the formation under sensing constraints in an obstacle-cluttered environment

[11]–[14] offer an additional transformation: isotropic geometric resizing, which significantly improves maneuverability.

However, existing scaling control methods are often inefficient in anisotropic settings. For example, in elongated corridors [2], uniform scaling may require unnecessary reduction along unconstrained directions, while non-uniform scaling enables selective compression along the constrained axis, better accommodating spatial or hardware constraints. In dynamic environments [15], non-uniform scaling further improves responsiveness by reducing superfluous transformations, which are often time-consuming. Although affine formation control [16]–[18] theoretically enables non-uniform scaling, existing methods face challenges due to complex sensing graph structures and the high computational cost of centralized optimization over constraint matrices. As a result, non-uniform scaling transformations in formation control remain relatively underexplored.

On the other hand, attitude formation control introduces additional complexity to maneuverability. Most existing approaches either seek full heading consensus, aligning all agents

to a common orientation for simplified coordination [19], or aim to maintain fixed relative attitudes to preserve structured formation patterns with constant orientation differences [9], [20], [21]. These approaches rarely account for the coupled nature of position and attitude in practical scenarios, and pay limited attention to scalable attitude adjustments that could significantly enhance the formation's agility and adaptability.

To address the above-mentioned challenges, we investigate distributed strategies for non-uniform scaling of formations, as demonstrated in Fig. 1. Moreover, we propose a novel distributed control framework that jointly regulates position and attitude formations.

The main contributions of this paper are as follows.

- 1) To ensure that all maneuver parameters are effective and the follower states are uniquely determined by the leader states, we introduce the concept of maximum maneuverability, and establish the necessary and sufficient graphical conditions under which a formation achieves maximum maneuverability within the leader-follower framework; see Section 3.
- 2) We design a local linear constraint and construct a matrix-valued Laplacian to characterize the target formation. An efficient method for computing the corresponding stabilizing matrix is developed (see Lemma VII.1 and Theorem IV.3). In contrast to existing approaches [11], [18], [22], where the stabilizing matrix is computed for the entire formation, our method enables decentralized computation over individual DEPs.
- 3) We propose a distributed non-uniform scaling maneuver control law for the joint position-attitude formation. Under a 2-rooted graph structure, we guarantee global convergence of the closed-loop system (see Theorems IV.1 and IV.2). Compared to existing affine formation maneuver control approaches [18], [23] that support non-uniform scaling, our method relies on a sparser sensing graph, requires fewer leaders, and additionally supports scaling transformations of the attitude formation.

The structure of this paper is organized as follows: Section 2 introduces the notations and formulates the problem. Section 3 presents the concept of maximum maneuverability and its necessary and sufficient conditions. Section 4 provides the controller design method. Section 5 includes simulations to validate the theoretical results. Section 6 concludes the paper.

II. PRELIMINARIES AND PROBLEM STATEMENT

A. Notations

Throughout this paper, $\mathbb{R}$ denotes the set of real numbers, $\mathbb{R}^d$ the d -dimensional Euclidean space, $\dim(\cdot)$ the dimension of a linear space, and $|\cdot|$ the cardinality of a set or the element-wise absolute value for a scalar, vector, or matrix. Let $\text{null}(\cdot)$, $\text{image}(\cdot)$, $\text{tr}(\cdot)$, $\det(\cdot)$ and $\text{rank}(\cdot)$ denote the null space, image space, trace, determinant and rank of a matrix, respectively. The identity matrix is $I_n \in \mathbb{R}^{n \times n}$, $\mathbf{1}_n \in \mathbb{R}^n$ the all-ones vector, $\mathbf{0}$ a zero tensor (scalar/vector/matrix) with context-appropriate dimensions, and $\otimes$ the Kronecker product.

For any vector $x = [x_1, \dots, x_d]^\top \in \mathbb{R}^d$, $\text{diag}(x)$ is the diagonal matrix with x_i as its i -th diagonal entry, and

$\text{diag}\{A_i\}$ the block-diagonal matrix with block A_i in its i -th diagonal position. The Special Orthogonal group is $\text{SO}(2) = \{R \in \mathbb{R}^{2 \times 2} : R^\top R = I_2, \det(R) = 1\}$, with $R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ as a rotation matrix. The Euclidean norm is $\|\cdot\|$, while $\wedge$, $\Rightarrow$, and $\Longleftrightarrow$ denote logical conjunction, implication, and equivalence, respectively.

B. Graph Theory

Consider a graph $G = (V, E)$ representing a multi-agent system, where the vertex set $V = \{1, \dots, n\}$ denotes the set of agents and the edge set $E \subseteq \{(i, k) : i, k \in V \text{ and } i \neq k\}$ captures sensing relationships. Each directed edge $(i, k) \in E$ indicates that agent k can measure information from agent i . We refer to G as a sensing graph since its edges explicitly encode the directional sensing relationships between agents. The neighbor set of agent k is defined as $N_k = \{i \in V : (i, k) \in E\}$.

The graph G is called bidirectional if, for every edge $(i, j) \in E$, its reverse edge (j, i) also belongs to E . A bidirectional path from agent i_1 to agent i_k is a sequence of distinct agents $i_1, i_2, \dots, i_k$ such that both $(i_l, i_{l+1}) \in E$ and $(i_{l+1}, i_l) \in E$ hold for all $l = 1, \dots, k-1$. The agents i_1 and i_k are called the end agents, while any intermediate agents are termed inner agents.

A matrix $M = [M_{ki}] \in \mathbb{R}^{n \times n}$ is called a matrix-valued Laplacian if $\sum_{i=1}^n M_{ki} = 0$ for $k = 1, \dots, n$, where the matrix $M_{ki} \in \mathbb{R}^{d \times d}$ corresponds to the directed edge $(i, k) \in E$.

We now introduce key definitions used throughout this work.

Definition II.1 ([11]). *For a bidirectional graph G , an agent i is said to be 2-reachable from a non-singleton set U of agents if there exists a bidirectional path from an agent in U to agent i after removing any agent except agent i .*

Definition II.2 ([11]). *A bidirectional graph G is said to be 2-rooted if there exists a set of two agents (called roots), from which every other agent is 2-reachable.*

Definition II.3 (Dual-Entry Path). *A dual-entry path (DEP) is a subgraph $G_{\mathcal{P}} = (V_{\mathcal{P}}, E_{\mathcal{P}})$ comprising two distinct entry agents $i, j \in V$ and an ordered sequence of $\ell \geq 1$ inner agents $1, \dots, \ell$ forming a bidirectional path, such that:*

- If $\ell = 1$, then $V_{\mathcal{P}} = \{i, j, 1\}$ and $E_{\mathcal{P}} = \{(i, 1), (j, 1)\}$;
- If $\ell \geq 2$, then $V_{\mathcal{P}} = \{i, j\} \cup \{1, \dots, \ell\}$ and $E_{\mathcal{P}} = \{(i, 1), (j, \ell)\} \cup \{(k, k+1), (k+1, k) : 1 \leq k < \ell\}$.

Definition II.4 (DEP-Induced Graph). *Let $\mathcal{L}_0 = (V_0, E_0)$ be the graph with agent set $V_0 = \{1, 2\}$ and edge set $E_0 = \emptyset$. For $h = 1, \dots, \kappa$, define $\mathcal{L}_h = (V_h, E_h)$ by attaching a DEP $G_{\mathcal{P}_h} = (V_{\mathcal{P}_h}, E_{\mathcal{P}_h})$ to $\mathcal{L}_{h-1} = (V_{h-1}, E_{h-1})$ via distinct entry agents $i_h, j_h \in V_{h-1}$, where $V_{\mathcal{P}_h} \cap V_{h-1} = \{i_h, j_h\}$, the inner agents of $G_{\mathcal{P}_h}$ are labeled as $\{|V_{h-1}| + 1, \dots, |V_{h-1}| + \ell_h\}$, and $V_h = V_{h-1} \cup V_{\mathcal{P}_h}$, $E_h = E_{h-1} \cup E_{\mathcal{P}_h}$.*

An example of dual-entry path induced graph (DEP-induced graph) is given in Fig. 2. The following result establishes a connection between 2-rooted graphs and the DEP-induced graph.

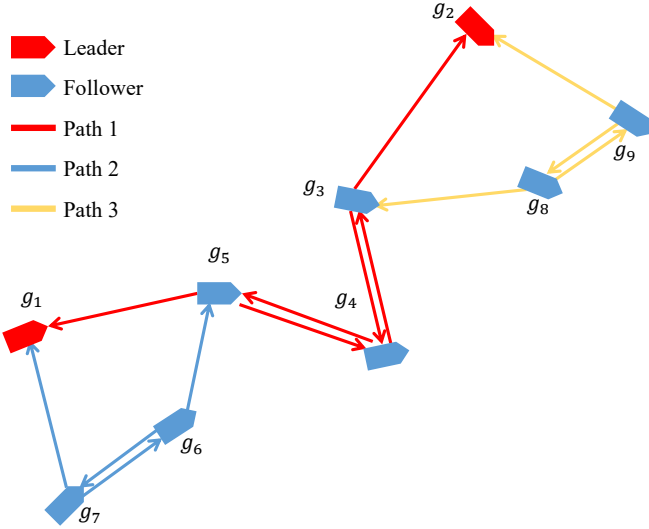


Fig. 2: A DEP-induced graph with three DEPs.

Lemma II.1. *A bidirectional graph G is 2-rooted if and only if it contains a DEP-induced graph as its spanning subgraph.*

Proof: See Appendix VII-A. ■

C. Affine Span and Diagonal Stability

This section establishes the geometric and algebraic foundations for formation stability.

Definition II.5 (Affine Span [18]). *The affine span of a set $\{x_i\}_{i=1}^n$ is defined by*

$$\mathcal{S}(\{x_i\}_{i=1}^n) = \left\{ \sum_{i=1}^n a_i x_i : a_i \in \mathbb{R}, \sum_{i=1}^n a_i = 1 \right\}. \quad (1)$$

By definition, it can be deduced that a set $\{x_i\}_{i=1}^n$ affinely spans $\mathbb{R}$ (i.e., $\mathcal{S}(\{x_i\}_{i=1}^n) = \mathbb{R}$) if $x_i \in \mathbb{R}, i = 1, \dots, n$, and $\text{rank}(\bar{P}(x)) = 2$, where

$$\bar{P}(x) = \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix}. \quad (2)$$

Equivalently, this holds if $n \geq 2$ and there exist at least two distinct x_i , i.e., $x_i \neq x_j$ for some $i \neq j$.

Lemma II.2 (Diagonal Stability [24], Theorem 3.2). *Let A be an $n \times n$ matrix whose all leading principal minors are nonzero. Then, there exists a diagonal matrix D such that every eigenvalue of DA has a positive real part.*

D. Joint Position-Attitude Formation

Consider a group of n agents in $\mathbb{R}^2$, the dynamics of the i -th agent is given by

$$\dot{g}_i = \begin{bmatrix} \dot{p}_i \\ \dot{\phi}_i \end{bmatrix} = \begin{bmatrix} u_i \\ \omega_i \end{bmatrix}, \quad (3)$$

where $g_i = [p_i^\top, \phi_i]^\top \in \mathbb{R}^3$, $p_i = [p_i^x, p_i^y]^\top \in \mathbb{R}^2$, $\phi_i \in \mathbb{R}$, $u_i \in \mathbb{R}^2$, and $\omega_i \in \mathbb{R}$ denote the state, position, yaw angle,

linear velocity and yaw rate of agent i in the world frame respectively.

A formation in $\mathbb{R}^2$, denoted by (G, g) , is defined as the combination of a configuration g and a sensing graph G . The configuration is given by the stacked state vector $g = [g_1^\top, \dots, g_i^\top, \dots, g_n^\top]^\top$.

By defining $p = [p_1^\top, \dots, p_i^\top, \dots, p_n^\top]^\top$ and $\phi = [\phi_1, \dots, \phi_i, \dots, \phi_n]^\top$, the formation is categorized depending on the type of configuration g as follows.

- If $g = p$, the formation is referred to as a position formation, denoted by (G, p) .
- If $g = \phi$, the formation is called an attitude formation, denoted by (G, ϕ) .
- If $g = [p_1^\top, \phi_1, \dots, p_n^\top, \phi_n]^\top$, the formation is termed a joint position-attitude formation, denoted by (G, g) .

As defined in [25], [26], position formation typically models agents as point masses, aiming to achieve a desired spatial configuration. In contrast, attitude formation focuses on the orientation of each agent, ensuring specific directional relationships—such as aligned or coordinated headings—among agents [27], [28]. To address more complex scenarios, this paper investigates a generalized joint position-attitude formation framework, where both the position and orientation of agents are simultaneously controlled. This approach enables finer regulation of the formation's global geometry and internal structure, extending the capabilities of traditional formation control strategies [9], [29], [30].

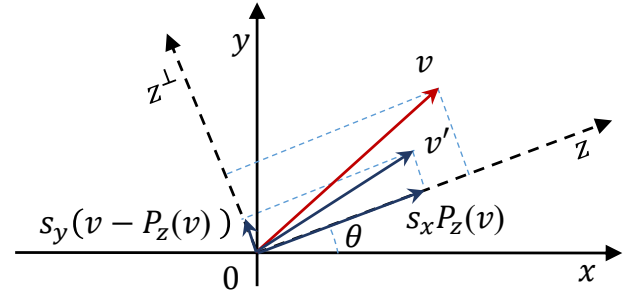


Fig. 3: Non-uniform scaling transformation in $z = [\cos \theta, \sin \theta]^\top$ direction.

E. Non-Uniform Scaling Transformation of Position Formation

Before defining the non-uniform scaling transformation for position formations, we first introduce the concept of non-uniform scaling transformation for a vector in $\mathbb{R}^2$.

As illustrated in Fig. 3, consider a non-uniform scaling transformation applied to a vector $v \in \mathbb{R}^2$ along an arbitrary direction $z = [\cos \theta, \sin \theta]^\top$, where θ is called the scaling direction. The transformation is characterized by directional scaling factors s_x and s_y , which correspond to the axis aligned with z and $z^\perp = R(\frac{\pi}{2})z$, respectively. The transformed vector

is given by:

$$\begin{aligned}
 v' &= s_x P_z(v) + s_y (v - P_z(v)) \\
 &= ((s_x - s_y) z z^\top + s_y I_2) v \\
 &= \left(R(\theta) \begin{bmatrix} s_x - s_y & 0 \\ 0 & 0 \end{bmatrix} R^\top(\theta) + s_y I_2 \right) v \\
 &= R(\theta) \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} R^\top(\theta) v.
 \end{aligned} \quad (4)$$

Here, $P_z(v) = z z^\top v$ denotes the projection of v onto direction z , and $R(\theta)$ is the rotation matrix aligning the x -axis with z . Note that when $s_x = s_y = s$, the transformation reduces to $v' = s v$, which corresponds to a uniform scaling case.

We now extend this concept to position formations.

Definition II.6 (Non-Uniform Scaling of Position Formation). *Given a nominal position formation $(G, \tilde{p})$ in $\mathbb{R}^2$ with configuration $\tilde{p} = [\tilde{p}_1^\top, \dots, \tilde{p}_i^\top, \dots, \tilde{p}_n^\top]^\top \in \mathbb{R}^{2n}$, its non-uniform scaling transformation associated with scaling direction θ is defined as:*

$$p' = (I_n \otimes (R(\theta) \text{diag}(s_p) R^\top(\theta))) \tilde{p}, \quad (5)$$

where $R(\theta) \in \text{SO}(2)$, $s_p = [s_x, s_y]^\top \in \mathbb{R}^2$ is the scaling factor vector.

This framework enables continuous modulation of formation shapes along arbitrary directions, providing a foundation for the anisotropic scaling formation maneuver control strategy proposed in this paper. Compared to uniform scaling methods [11], [25], [31] and fixed scaling approaches [6], [7], [10], the proposed non-uniform scaling offers superior flexibility in controlling multi-agent formations.

F. Scaling and Translation Transformation of Attitude Formation

Existing approaches to attitude formation control primarily address either consensus alignment [19] or fixed relative attitudes [9], [20], [21]. While these methods enable basic coordination patterns, their limited adaptability restricts their capacity to meet dynamic operational requirements. To overcome this limitation, we propose a framework for scaling and translation transformations in attitude formation, which enables continuous modulation of formation geometry through scaling and translation operations. This subsection provides a detailed definition of these transformations:

Definition II.7 (Scaling and Translation of Attitude Formation). *Given a nominal attitude formation $(G, \tilde{\phi})$ with configuration $\tilde{\phi} = [\tilde{\phi}_1, \dots, \tilde{\phi}_i, \dots, \tilde{\phi}_n]^\top \in \mathbb{R}^n$, its scaling and translation transformation is defined as:*

$$\phi' = (I_n \otimes s_\phi) \tilde{\phi} + 1_n \otimes \tau_\phi, \quad (6)$$

where $s_\phi \in \mathbb{R}$ and $\tau_\phi \in \mathbb{R}$ are the scaling and translation factors respectively.

Three examples of attitude formation transformations are given in Fig. 4 to demonstrate Definition II.7, where each arrow represents the yaw angle of an agent. Detailed explanations for the two types of transformations are given below, respectively.

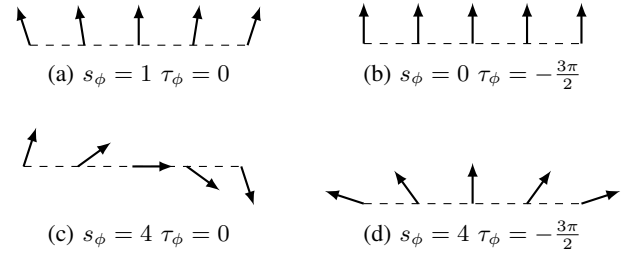


Fig. 4: Attitude formation transformation. (a) Original formation $\phi = [\frac{3\pi}{5}, \frac{11\pi}{20}, \frac{\pi}{2}, \frac{9\pi}{20}, \frac{2\pi}{5}]$. (b) Translation only. (c) Scaling only. (d) Scaling + translation.

The scaling factor s_ϕ modulates the relative differences in yaw angles between agents. If $|s_\phi| > 1$, the relative yaw angles are amplified, resulting in a more “divergent” orientation structure among the agents. Conversely, $0 < |s_\phi| < 1$ compresses the differences in orientation, making the agents more aligned. This transformation allows the adjustment of the relative angular dispersion within the formation.

In contrast, the translation factor τ_ϕ can be interpreted as a uniform offset applied to all agents’ yaw angles. Geometrically, this corresponds to each agent rotating around its own center by the same angle τ_ϕ . This transformation preserves the relative orientation between agents and results in a rigid-body rotation of the entire formation in the attitude space.

G. Problem Statement

In this article, we aim to achieve combined transformations including translation, and non-uniform scaling of the nominal joint position-attitude formation by tuning only the states of partial agents. As shown in Fig. 1, when avoiding obstacles, a formation that can perform a non-uniform scaling transformation in an arbitrary direction is more environmentally friendly and efficient compared to those that can only perform uniform scaling transformation in the literature [32] [33].

We adopt a leader-follower strategy for formation maneuver control. Consider a formation comprising m leaders and $n - m$ followers, with the leader set denoted as $V_l = \{1, \dots, m\}$ and the follower set as $V_f = \{m + 1, \dots, n\}$. The states for the leaders and followers are defined as $g_l = [g_1^\top, \dots, g_m^\top]^\top \in \mathbb{R}^{3m}$ and $g_f = [g_{m+1}^\top, \dots, g_n^\top]^\top \in \mathbb{R}^{3(n-m)}$, respectively.

1) *Target Formation:* We focus on the nominal joint position-attitude formation subject to non-uniform scaling along a specified direction. To explicitly represent such a setting, we extend the formation representation from the pair (G, g) to a triple $(G, \tilde{g}, \theta)$, where G remains the underlying sensing graph, while $(\tilde{g}, \theta)$ jointly describes an arbitrarily chosen nominal configuration for the team of agents. Specifically, the nominal state $\tilde{g} = [\tilde{g}_l^\top, \tilde{g}_f^\top]^\top$, where $\tilde{g}_l = [\tilde{g}_1^\top, \dots, \tilde{g}_m^\top]^\top \in \mathbb{R}^{3m}$ represents the nominal state corresponding to the leaders, and $\tilde{g}_f = [\tilde{g}_{m+1}^\top, \dots, \tilde{g}_n^\top]^\top \in \mathbb{R}^{3(n-m)}$ denotes the nominal state for the followers. Each component $\tilde{g}_i = [\tilde{p}_i^\top, \tilde{\phi}_i]^\top \in \mathbb{R}^3$ consists of $\tilde{p}_i = [\tilde{p}_i^x, \tilde{p}_i^y]^\top \in \mathbb{R}^2$ and $\tilde{\phi}_i \in \mathbb{R}$. Furthermore, θ is the nominal scaling direction.

The time-varying target state of $(G, \tilde{g}, \theta)$ is parameterized by the stacked vector $g^*(t) = [g_l^{*\top}(t), g_f^{*\top}(t)]^\top$, where

$g_l^*(t) = [g_1^{*\top}(t), \dots, g_m^{*\top}(t)]^\top \in \mathbb{R}^{3m}$ and $g_f^*(t) = [g_{m+1}^{*\top}(t), \dots, g_n^{*\top}(t)]^\top \in \mathbb{R}^{3(n-m)}$ represent the target states for the leaders and followers, respectively. These target states evolve continuously over time with reference to the nominal configuration $(\tilde{g}, \theta)$. Specifically:

$$g^*(t) = (I_n \otimes S(t, \theta))\tilde{g} + 1_n \otimes \tau(t), \quad (7)$$

where

$$S(t, \theta) = \begin{bmatrix} R(\theta) \text{diag}(s_p(t))R^\top(\theta) & 0 \\ 0 & s_\phi(t) \end{bmatrix} \\ = \Theta \text{diag}(s(t))\Theta^\top \in \mathbb{R}^{3 \times 3},$$

t is the time variable, $R(\theta) \in \text{SO}(2)$, $\Theta = \begin{bmatrix} R(\theta) & 0 \\ 0 & 1 \end{bmatrix}$, $s(t) = [s_p^\top(t), s_\phi(t)]^\top \in \mathbb{R}^3$ and $\tau(t) = [\tau_p^\top(t), \tau_\phi(t)]^\top \in \mathbb{R}^3$ are time-varying maneuver parameters corresponding to the joint position-attitude formation, where:

- $s_p(t) \in \mathbb{R}^2$ governs the non-uniform scaling of the position formation along the axes of a frame defined by the scaling direction $\theta \in \mathbb{R}$ as defined in Definition II.6, and $\tau_p(t) = [\tau_x(t), \tau_y(t)]^\top \in \mathbb{R}^2$ is the time-varying translation maneuver parameter of the position formation;
- $s_\phi(t) \in \mathbb{R}$ and $\tau_\phi(t) \in \mathbb{R}$ determine the scaling and translation of the attitude formation, respectively, as specified in Definition II.7.

2) *Sensing Capability*: Each follower agent is not able to communicate with others, and can only access local relative measurements, including: (i) the relative positions $\{p_j - p_i\}_{j \in N_i}$ and (ii) relative yaw angles $\{\phi_j - \phi_i\}_{j \in N_i}$.

Each leader agent, functioning as a mobile reference, has the enhanced capability of measuring its absolute state within the global coordinate frame.

This heterogeneous sensing paradigm aligns with practical scenarios, where leaders may carry high-precision sensors (e.g., IMU-GPS fusion systems [34]) while followers rely on onboard vision, UWB or LiDAR for local observations [35]–[37].

The distributed non-uniform scaling formation maneuver control problem is then defined as follows.

Problem II.1 (Non-Uniform Scaling Formation Maneuver Control). *Given a nominal configuration $(\tilde{g}, \theta)$ known to all agents, and the desired time-varying maneuver parameters $s(t)$, $\tau(t)$ only available to leaders, design a distributed controller (u_i, ω_i) based on local measurements, such that all the agents, subject to (3), achieve the following objective:*

$$\lim_{t \rightarrow \infty} (g_i(t) - g_i^*(t)) = 0, i \in V, \quad (8)$$

where $g^*(t) = [\dots, g_i^{*\top}(t), \dots]^\top$ is determined by $(\tilde{g}, \theta)$ and the maneuver parameters according to (7).

III. MAXIMUM MANEUVERABILITY AND MATRIX-VALUED LAPLACIAN

To solve the distributed non-uniform scaling formation maneuver control problem described in Problem II.1, we first analyze the conditions on the nominal configuration that ensure all maneuver parameters are effective. Next, we investigate

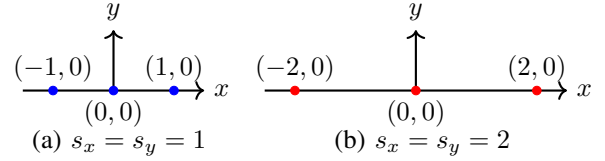


Fig. 5: Singular configuration. (a) Original positions of the agents are aligned with the x-axis. (b) Under scaling transformation with $\theta = 0$, see Equation (5), a structural singularity occurs: the y-axis scaling parameter s_y becomes ineffective, while x-axis scaling remains effective.

how to select leaders and design formation rules so that the leaders can fully govern the behavior of the followers, thereby achieving maximal control over the formation (we refer to this system-wide property as maximum maneuverability). Finally, we derive the rank and graph conditions required for maximum maneuverability.

A. Maximum Maneuverability

In reality, certain nominal configurations can introduce singularities that undermine the effectiveness of maneuver parameters. As shown in Fig. 5, when all agents are aligned along the x-axis, scaling along the y-axis has no effect on the formation geometry, while x-axis scaling remains effective. In this case, the y-axis scaling parameter s_y becomes ineffective, resulting in limited maneuverability and inapplicability to complex tasks, such as transitioning from a line to a V-shape. Next, we formalize the concept of a non-singular configuration to address this issue.

From (7), given θ and $\tilde{g}$, the time-varying target state $g^*(t)$ varies with the maneuver parameters, and all possible states form a space $\Pi(\tilde{g}, \theta)$. We term $\Pi(\tilde{g}, \theta)$ as the target state space, as defined by the following equation:

$$\begin{aligned} \Pi(\tilde{g}, \theta) &= \{g \in \mathbb{R}^{3n} : g = (I_n \otimes S(\theta))\tilde{g} + 1_n \otimes \tau, \\ &\quad S(\theta) = \Theta \text{diag}(s)\Theta^\top, s, \tau \in \mathbb{R}^3\} \\ &= \{g \in \mathbb{R}^{3n} : g_i = \tau + \Theta \text{diag}(s)\Theta^\top \tilde{g}_i = \tau + \\ &\quad \Theta \text{diag}(\Theta^\top \tilde{g}_i)s, s, \tau \in \mathbb{R}^3, i = 1, \dots, n\} \\ &= \{g \in \mathbb{R}^{3n} : g = \mathcal{F}(s, \tau), s, \tau \in \mathbb{R}^3\}, \end{aligned} \quad (9)$$

where $\mathcal{F}(s, \tau) \triangleq A(\tilde{g}, \theta)[s^\top, \tau^\top]^\top$, $\tilde{g}_{i, \theta} = \Theta^\top \tilde{g}_i = [\tilde{p}_{i, \theta}^x, \tilde{p}_{i, \theta}^y, \tilde{\phi}_i]^\top$,

$$A(\tilde{g}, \theta) = \begin{bmatrix} \Theta \text{diag}(\tilde{g}_{1, \theta}) & I_3 \\ \vdots & \vdots \\ \Theta \text{diag}(\tilde{g}_{n, \theta}) & I_3 \end{bmatrix} \in \mathbb{R}^{3n \times 6}. \quad (10)$$

Definition III.1 (Non-Singular Configuration). *A nominal configuration $(\tilde{g}, \theta)$ is non-singular if the mapping $\mathcal{F}$ is injective, and is singular otherwise.*

By the above definition, a non-singular configuration ensures that all maneuver parameters uniquely determine the target state. Next, we establish equivalent conditions for a non-singular configuration.

Lemma III.1. *The following statements are equivalent:*

- (a) $(\tilde{g}, \theta)$ is non-singular;
- (b) $\text{rank}(A(\tilde{g}, \theta)) = 6$;
- (c) $\dim(\Pi(\tilde{g}, \theta)) = 6$;
- (d) each of the sets $\{\tilde{p}_{i,\theta}^x\}_{i \in V}$, $\{\tilde{p}_{i,\theta}^y\}_{i \in V}$, and $\{\tilde{\phi}_i\}_{i \in V}$ affinely spans $\mathbb{R}$.

Proof: See Appendix VII-B. ■

It is worth noting that the three translation maneuver parameters τ_x , τ_y , and τ_ϕ are always effective. In contrast, the effectiveness of the three scaling maneuver parameters requires the validity of the three conditions in Lemma III.1(d). For example, if the set $\{\tilde{p}_{i,\theta}^x\}_{i \in V}$ does not affinely span $\mathbb{R}$, then the maneuver parameter s_x becomes ineffective.

Lemma III.1 motivates our core assumption about the nominal configuration as follows.

Assumption III.1. For a nominal formation $(G, \tilde{g}, \theta)$ in $\mathbb{R}^2$, each of the sets $\{\tilde{p}_{i,\theta}^x\}_{i \in V}$, $\{\tilde{p}_{i,\theta}^y\}_{i \in V}$, and $\{\tilde{\phi}_i\}_{i \in V}$ affinely spans $\mathbb{R}$.

To enable formation maneuver control with robust adaptability to complex environments and diverse mission requirements, maintaining maximum maneuverability is essential. Under the leader-follower strategy, a singular nominal configuration of the leaders compromises formation maneuverability by rendering certain maneuver parameters ineffective.

Moreover, even if the leader configuration is non-singular, followers constrained by local sensing may still fail to track leader state changes. This highlights the challenge of ensuring that the influence of leader motions can fully and uniquely propagate throughout the formation. Inspired by [18], [26], [38]–[44], we seek a Laplacian $M = [M_{ki}] \in \mathbb{R}^{3n \times 3n}$ determined by $(G, \tilde{g}, \theta)$ such that

$$\Pi(\tilde{g}, \theta) = \{g \in \mathbb{R}^{3n} : Mg = 0\}, \quad (11)$$

where $g = [g_l^\top, g_f^\top]^\top$ represents the combined state of leaders and followers, and M_{ki} , defined based on the local measurement of agent k , reflects the interaction weight between agent k and agent i within the formation constraints. If the follower states g_f are uniquely determined by the leader states g_l through $Mg = 0$, any change in the leader states induces a corresponding change in the follower states.

Now, we formally define maximum maneuverability in the leader-follower framework as follows.

Definition III.2. A nominal formation $(G, \tilde{g}, \theta)$ in $\mathbb{R}^2$ achieves maximum maneuverability under the leader-follower strategy with Laplacian M if

- (a) the leaders' nominal configuration $(\tilde{g}_l, \theta)$ is non-singular;
- (b) for any $g = [g_l^\top, g_f^\top]^\top \in \Pi(\tilde{g}, \theta)$, the follower state g_f is uniquely determined by the leader state g_l through the constraint $Mg = 0$.

B. Leader Selection for Maximum Maneuverability

The following lemma further gives equivalent conditions for the convenience of leader selection.

Lemma III.2 (Leader Selection for Maximum Maneuverability). The leaders' nominal configuration $(\tilde{g}_l, \theta)$ is non-singular if and only if the following conditions are satisfied:

- (a) the number of leaders satisfies $m \geq 2$;
- (b) each of the sets $\{\tilde{p}_{i,\theta}^x\}_{i \in V_l}$, $\{\tilde{p}_{i,\theta}^y\}_{i \in V_l}$, and $\{\tilde{\phi}_i\}_{i \in V_l}$ affinely spans $\mathbb{R}$.

Proof: According to Lemma III.1, the result follows directly. ■

When the leader nominal configuration is non-singular, there exists a one-to-one correspondence between the leaders' states and the maneuver parameters. Next we show how to explicitly compute these maneuver parameters.

From (9), we have $g_l = A(\tilde{g}_l, \theta)z$, where $z = [s^\top, \tau^\top]^\top$, and $A(\tilde{g}_l, \theta) \in \mathbb{R}^{3m \times 6}$ is obtained by substituting $\tilde{g}$ in Equation (10) with $\tilde{g}_l$. Lemma III.2 implies that $\text{rank}(A(\tilde{g}_l, \theta)) = 6$. As a result, the maneuver parameters can be uniquely determined as:

$$z = (A^\top(\tilde{g}_l, \theta)A(\tilde{g}_l, \theta))^{-1} A^\top(\tilde{g}_l, \theta)g_l. \quad (12)$$

C. Matrix-Valued Laplacian for Maximum Maneuverability

To construct a Laplacian matrix satisfying Definition III.2, we firstly introduce the following matrix-valued constraint involving three agents i, j, k :

$$W_{jk}(\tilde{g}_{ijk}, \theta)g_{ik} + W_{ki}(\tilde{g}_{ijk}, \theta)g_{jk} = 0, \quad (13)$$

where $g_{ik} = g_i - g_k$, $g_{jk} = g_j - g_k$, $W_{jk}(\tilde{g}_{ijk}, \theta) = w_{jk}\Theta^\top$, $W_{ki}(\tilde{g}_{ijk}, \theta) = w_{ki}\Theta^\top$, $\Theta = \begin{bmatrix} R^\top(\theta) & 0 \\ 0 & 1 \end{bmatrix}$, $w_{ki} = \begin{bmatrix} \text{diag}(\tilde{p}_{ki,\theta}) & 0 \\ 0 & \tilde{\phi}_{ki} \end{bmatrix}$, $w_{jk} = \begin{bmatrix} \text{diag}(\tilde{p}_{jk,\theta}) & 0 \\ 0 & \tilde{\phi}_{jk} \end{bmatrix}$, $\tilde{g}_{ijk} = [\tilde{g}_i^\top, \tilde{g}_j^\top, \tilde{g}_k^\top]^\top$.

Taking Fig. 6 as an example, the states of agents i, j, k are $g_i = [-2, 1, \pi/2]^\top$, $g_j = [1.5, 0.5, \pi/4]^\top$, $g_k = [0, 0, 0]^\top$, respectively, and the scaling direction is $\theta = 0$. Then,

$$W_{jk} = \begin{bmatrix} 1.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & \pi/4 \end{bmatrix}, \quad W_{ki} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -\pi/2 \end{bmatrix}. \quad (14)$$

The constant-value matrices apply a non-uniform scaling transformation to the relative state vector, ensuring that the sum of two directed edges under this transformation equals zero. Note that the choice of constant-value matrices is not unique.

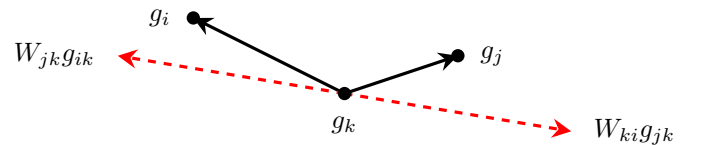


Fig. 6: An example of the matrix-valued constraint.

We now present a key property of the matrix-valued constraint.

Lemma III.3. The constraint (13) is invariant to translation and non-uniform scaling transformation of g_i, g_j, g_k .

Proof: For each $l \in \{i, j, k\}$, we apply a translation τ and a non-uniform scaling transformation S , obtaining:

$$g'_l = Sg_l + \tau, \quad l \in \{i, j, k\}. \quad (15)$$

We now demonstrate that the transformed vectors g'_i, g'_j, g'_k satisfy the matrix-valued constraint in (13):

$$\begin{aligned} W_{jk}g'_{ik} + W_{ki}g'_{jk} &= W_{jk}Sg_{ik} + W_{ki}Sg_{jk} \\ &= \begin{bmatrix} \text{diag}(\tilde{p}_{jk,\theta}) & 0 \\ 0 & \tilde{\phi}_{jk} \end{bmatrix} \text{diag}(s)\Theta^\top g_{ik} \\ &\quad + \begin{bmatrix} \text{diag}(\tilde{p}_{ki,\theta}) & 0 \\ 0 & \tilde{\phi}_{ki} \end{bmatrix} \text{diag}(s)\Theta^\top g_{jk} \\ &= \text{diag}(s)(W_{jk}g_{ik} + W_{ki}g_{jk}) = 0, \end{aligned} \quad (16)$$

the final equality holds because multiplication of diagonal matrices is commutative. ■

Next, we construct a matrix-valued Laplacian based on the proposed matrix-valued constraint. Let the constraint index set be defined as $C = \{(i, j, k) \in V^3 : (i, k), (j, k) \in E, i < j\}$. The set of all constraints associated with the sensing graph G is then given by $\{W_{jk}g_{ik} + W_{ki}g_{jk} = 0 : (i, j, k) \in C\}$.

Each matrix-valued constraint defined in (13) corresponding to the constraint index $(i, j, k) \in C$ can be aggregated into a matrix-valued Laplacian $M(G, \tilde{g}, \theta) \in \mathbb{R}^{3n \times 3n}$ satisfying:

$$Mg = 0, \quad (17)$$

where the matrix block located at the k th row and i th column of M , denoted M_{ki} , is defined as follows:

$$M_{ki} = \begin{cases} \sum_{(i,j,k) \in C} W_{jk} + \sum_{(j',i,k) \in C} W_{kj'}, & \text{if } k \neq i, \\ \sum_{(i,j,k) \in C} W_{ij} + \sum_{(j',i,k) \in C} W_{j'i}, & \text{if } k = i. \end{cases} \quad (18)$$

By convention, each summation is defined to be zero when the corresponding index set is empty.

We now investigate key properties of the matrix-valued Laplacian $M(G, \tilde{g}, \theta)$.

Lemma III.4. *For any nominal configuration $(\tilde{g}, \theta)$, it always holds that $\Pi(\tilde{g}, \theta) \subseteq \text{null}(M(G, \tilde{g}, \theta))$.*

Proof: By the definition of the matrix-valued Laplacian $M(G, \tilde{g}, \theta)$ and Lemma III.3, the result follows directly. ■

Lemma III.5. *Under Assumption III.1, the following conditions are equivalent:*

- (a) $\text{null}(M(G, \tilde{g}, \theta)) = \Pi(\tilde{g}, \theta)$,
- (b) $\text{rank}(M(G, \tilde{g}, \theta)) = 3n - 6$.

Proof: From Lemma III.4, we have $\Pi(\tilde{g}, \theta) \subseteq \text{null}(M(G, \tilde{g}, \theta))$. Thus, $\text{null}(M(G, \tilde{g}, \theta)) = \Pi(\tilde{g}, \theta)$ if and only if $\dim(\text{null}(M(G, \tilde{g}, \theta))) = \dim(\Pi(\tilde{g}, \theta))$. Given that $\dim(\Pi(\tilde{g}, \theta)) = 6$ under Assumption III.1, it follows that $\text{null}(M(G, \tilde{g}, \theta)) = \Pi(\tilde{g}, \theta)$ if and only if $\text{rank}(M(G, \tilde{g}, \theta)) = 3n - 6$. ■

To this point, we have derived the condition on the matrix-valued Laplacian for characterizing the target configuration space $\Pi(\tilde{g}, \theta)$. Next, we investigate how the leader states

uniquely determine the follower states through the matrix-valued Laplacian. We begin by reformulating (17) as

$$Mg = \hat{M} \text{diag}(\Theta^\top)g = 0, \quad (19)$$

and subsequently partition $\hat{M}$ based on the leader-follower structure to facilitate this analysis.

$$\hat{M} = \begin{bmatrix} \hat{M}_l \\ \hat{M}_f \end{bmatrix} = \begin{bmatrix} \hat{M}_{ll} & \hat{M}_{lf} \\ \hat{M}_{fl} & \hat{M}_{ff} \end{bmatrix}, \quad (20)$$

where $\hat{M}_l = [\hat{M}_{ll} \ \hat{M}_{lf}]$, $\hat{M}_f = [\hat{M}_{fl} \ \hat{M}_{ff}]$, $\hat{M}_{ll} \in \mathbb{R}^{3m \times 3m}$, $\hat{M}_{lf} \in \mathbb{R}^{3m \times 3(n-m)}$, $\hat{M}_{fl} \in \mathbb{R}^{3(n-m) \times 3m}$ and $\hat{M}_{ff} \in \mathbb{R}^{3(n-m) \times 3(n-m)}$. Based on this partitioning, we obtain

$$\hat{M}_{fl} \text{diag}(\Theta^\top)g_l + \hat{M}_{ff} \text{diag}(\Theta^\top)g_f = 0. \quad (21)$$

If the block matrix $\hat{M}_{ff}$ is non-singular, the follower state g_f can be uniquely determined by

$$g_f = -\text{diag}(\Theta)\hat{M}_{ff}^{-1}\hat{M}_{fl} \text{diag}(\Theta^\top)g_l. \quad (22)$$

Based on this analysis and Lemma III.2, Definition III.2 can be reformulated as follows:

Definition III.3. *A nominal formation $(G, \tilde{g}, \theta)$ in $\mathbb{R}^2$ achieves maximum maneuverability under leader-follower strategy if and only if the following conditions are satisfied:*

- (a) *The number of leaders satisfies $m \geq 2$;*
- (b) *The sets $\{\tilde{p}_{i,\theta}^x\}_{i \in V_l}$, $\{\tilde{p}_{i,\theta}^y\}_{i \in V_l}$, and $\{\tilde{\phi}_i\}_{i \in V_l}$ each affinely span $\mathbb{R}$;*
- (c) *The block matrix $\hat{M}_{ff}$ in (21) is non-singular.*

D. Sensing Graphs for Maximum Maneuverability

According to Definition III.3, the non-singularity of the block matrix $\hat{M}_{ff}$ in (21) is a prerequisite for the formation $(G, \tilde{g}, \theta)$ to achieve maximum maneuverability. In what follows, we establish the necessary and sufficient conditions under which $\hat{M}_{ff}$ is non-singular. These conditions are associated with both the topological structure of the bidirectional sensing graph G and the nominal configuration $(\tilde{g}, \theta)$.

Lemma II.1 enables us to characterize the non-singularity of $\hat{M}_{ff}$ by imposing suitable non-degeneracy conditions on the configuration $(\tilde{g}, \theta)$ along each DEP. The formal condition is stated below.

Assumption III.2. *Consider a nominal formation $(G, \tilde{g}, \theta)$ in $\mathbb{R}^2$, where G is a 2-rooted graph with a spanning DEP-induced graph $\mathcal{L}_\kappa$, each DEP $G_{\mathcal{P}_h} = (V_{\mathcal{P}_h}, E_{\mathcal{P}_h})$, $h = 1, \dots, \kappa$, with entry agents $\{i_h, j_h\}$, satisfies:*

$$\prod_{\{u,v\} \in E_u} \tilde{p}_{uv,\theta}^x \tilde{p}_{uv,\theta}^y \tilde{\phi}_{uv} \neq 0, \quad (23)$$

where $E_u = \{\{u, v\} : (u, v), (v, u) \in E_{\mathcal{P}_h}\} \cup \{i_h, j_h\}$, $\tilde{p}_{uv,\theta}^x = \tilde{p}_{u,\theta}^x - \tilde{p}_{v,\theta}^x$, $\tilde{p}_{uv,\theta}^y = \tilde{p}_{u,\theta}^y - \tilde{p}_{v,\theta}^y$, and $\tilde{\phi}_{uv} = \phi_u - \phi_v$.

Equation (23) implies that the entry pair $\{i_h, j_h\}$ and all bidirectional edges $(u, v) \in E_{\mathcal{P}_h}$ satisfy that the projected nominal position differences $\tilde{p}_{uv,\theta}^x$, $\tilde{p}_{uv,\theta}^y$, and the relative nominal orientation $\tilde{\phi}_{uv}$ are all nonzero. This means that in the nominal configuration, no edge is parallel to the x-axis

or y-axis, and each pair of neighboring agents have different headings. Such settings ensure that all maneuver parameters are effective and can propagate through the formation.

We now give a graphical condition for maximum maneuverability.

Theorem III.1. *A nominal formation $(G, \tilde{g}, \theta)$ in $\mathbb{R}^2$ achieves maximum maneuverability under a leader-follower strategy if and only if G is 2-rooted with the two roots as leaders, and the nominal formation $(G, \tilde{g}, \theta)$ satisfies Assumption III.2.*

Proof: See Appendix VII-C. ■

Theorem III.1 provides a necessary and sufficient condition in terms of the sensing graph and the nominal configuration for maximum maneuverability. In practice, both the graph G and the nominal configuration $(\tilde{g}, \theta)$ can be artificially designed to satisfy the conditions.

Remark III.1. *Previous works on affine formation maneuver control (e.g., [22]) typically require a rank condition on the Laplacian, rather than looking at the graph structure. Although [11] introduced graphical conditions, they did not provide an explicit characterization of the infeasible nominal configurations. In contrast, our approach utilizes the DEP-induced graph to explicitly relate 2-rooted structures to maximal maneuverability. Moreover, the non-degeneracy condition (23) precisely characterizes the required geometric constraints.*

IV. NON-UNIFORM SCALING FORMATION MANEUVER CONTROL

Based on the preceding analysis, we propose the designed distributed controller in this section.

A. Distributed Formation Maneuver Control Laws

In this subsection, we propose distributed non-uniform scaling formation maneuver control laws, in the scenarios with stationary leaders and moving leaders, respectively.

According to the control objective described in Problem II.1, we define the tracking errors for followers and leaders as $\delta_l = g_l - g_l^*$ and $\delta_f = g_f - g_f^*$, respectively, where $g_f^* = (\hat{M}_{ff} \text{diag}(\Theta^\top))^{-1} \hat{M}_{fl} \text{diag}(\Theta^\top) g_l^* + g_f$. The control objective is thus reformulated as designing a distributed control law such that $\delta_f \rightarrow 0$ and $\delta_l \rightarrow 0$ as $t \rightarrow \infty$.

1) *Stationary Leaders:* We first consider the case where leaders are stationary, i.e., $g_l = g_l^*$ and $\dot{g}_l^* = 0$. In this case, the compact form of the formation control law is given by

$$\begin{cases} \dot{g}_l = 0, \\ \dot{g}_f = -\text{diag}(\Theta) D \hat{M}_{ff} \text{diag}(\Theta^\top) \delta_f, \end{cases} \quad (24)$$

where $D = \text{diag}(D_k)$ is a diagonal matrix to be designed to ensure the convergence of the tracking error, and each $D_k \in \mathbb{R}^{3 \times 3}$ is a non-zero diagonal gain matrix corresponding to agent k .

According to (18), the formation controller of each follower can be written as

$$\dot{g}_k = -\Theta D_k \sum_{(i,j,k) \in C} (W_{jk} g_{ik} + W_{ki} g_{jk}), k \in V_f. \quad (25)$$

The explicit form of (25) reveals that the controller of each individual agent relies solely on the relative state measurements of its neighbors. To guarantee the stability of the controller, the following assumption is made.

Assumption IV.1. *Consider a nominal formation $(G, \tilde{g}, \theta)$ in $\mathbb{R}^2$, where G is a 2-rooted graph with a spanning DEP-induced graph $\mathcal{L}_\kappa$. Each DEP $G_{\mathcal{P}_h} = (V_{\mathcal{P}_h}, E_{\mathcal{P}_h})$, $h = 1, \dots, \kappa$, with entry agents $\{i_h, j_h\}$ and inner agents $\{1, \dots, \ell_h\}$, satisfies:*

$$\prod_{l=2}^{\ell_h} \tilde{p}_{i_h l, \theta}^x \tilde{p}_{i_h l, \theta}^y \tilde{\phi}_{i_h l} \neq 0, \quad (26)$$

This assumption implies that for each DEP $G_{\mathcal{P}_h}$, all inner agents $l = 2, \dots, \ell_h$ must satisfy that the projected nominal position differences $\tilde{p}_{i_h l, \theta}^x$, $\tilde{p}_{i_h l, \theta}^y$, and the relative nominal orientation $\tilde{\phi}_{i_h l}$ with respect to the entry agent i_h are all nonzero. This means that no inner agent is horizontally or vertically aligned with i_h , and no inner agents share the same heading with i_h . This condition is generically satisfied, failing only on a measure zero subset of configurations.

Theorem IV.1. *Let the nominal formation $(G, \tilde{g}, \theta)$ in $\mathbb{R}^2$ satisfy Assumptions III.2 and IV.1. There exists a diagonal matrix D such that the tracking error δ_f converges to zero globally and exponentially fast under the control law (24).*

Proof: Substituting (24) into $\dot{\delta}_f$ gives

$$\begin{aligned} \dot{\delta}_f &= (\hat{M}_{ff} \text{diag}(\Theta^\top))^{-1} \hat{M}_{fl} \text{diag}(\Theta^\top) \dot{g}_l^* + \dot{g}_f \\ &= -\text{diag}(\Theta) D \hat{M}_{ff} \text{diag}(\Theta^\top) \delta_f. \end{aligned} \quad (27)$$

We first establish that under Assumptions III.2 and IV.1, there exists a diagonal matrix D such that every eigenvalue of $D \hat{M}_{ff}$ has a positive real part.

From equation (50), we observe that the spectrum of $\hat{M}_{ff}$ is determined by its block diagonal components $\hat{M}_{ff}^h$ (where $h \in 1, 2, \dots, \kappa$), each corresponding to the DEP graph $G_{\mathcal{P}_h}$. Under Assumptions III.2 and IV.1, Lemma VII.1 guarantees that for each diagonal block $\hat{M}_{ff}^h$ (corresponding to path graph $G_{\mathcal{P}_h}$), there exists a diagonal D^h such that $\sigma(D^h \hat{M}_{ff}^h)$ consists of eigenvalues with positive real parts, where $\sigma(\cdot)$ denotes the matrix spectrum.

Taking $D = \text{diag}\{D^h\}$ yields $D \hat{M}_{ff}$ with spectrum $\bigcup_{k=1}^{\kappa} \sigma(D^h \hat{M}_{ff}^h)$. Since all eigenvalues within each block have positive real parts, and blocks correspond to different path graphs, the combined spectrum maintains these properties.

Next, since $\text{diag}(\Theta)$ is non-singular, the matrices $\text{diag}(\Theta) D \hat{M}_{ff} \text{diag}(\Theta^\top)$ and $D \hat{M}_{ff}$ share the same eigenvalues. Consequently, all eigenvalues of $-\text{diag}(\Theta) D \hat{M}_{ff} \text{diag}(\Theta^\top)$ lie in the open left half plane. This implies that the tracking error δ_f converges to zero globally and exponentially. ■

Remark IV.1. *In [11], the authors showed that a stabilizing matrix exists for almost all Laplacians with a kernel space containing the nominal configuration. However, the infeasible cases are not clearly given. In contrast, we propose Assumption IV.1 as an explicit condition on the nominal configuration, under which the existence of a stabilizing matrix can always*

be guaranteed if the Laplacian matrix is designed according to (18).

2) *Moving Leaders*: To address moving leaders with time-varying velocities, we propose a formation maneuver control law that utilizes absolute velocity feedback, similar to the approach in [18], [45].

$$\dot{g}_k = \begin{cases} -k_l(g_k - g_k^*) + \dot{g}_k^*, & k \in V_l, \\ W_{kk}^{-1}[W_{jk}(k_f g_{ik} + \dot{g}_i) + W_{ki}(k_f g_{jk} + \dot{g}_j)], & k \in V_f, \end{cases} \quad (28)$$

where $W_{kk} = W_{jk} + W_{ki}$, $k_l \in \mathbb{R}$ and $k_f \in \mathbb{R}$ are positive control gains.

To guarantee the stability of the controller, the following assumption is required.

Assumption IV.2. For each matrix-valued constraint $(i, j, k) \in C$ defined in (13), the nominal configuration $(\tilde{g}, \theta)$ satisfies $\tilde{p}_{ji, \theta}^x \neq 0$, $\tilde{p}_{ji, \theta}^y \neq 0$, and $\tilde{\phi}_{ji} \neq 0$.

This assumption implies that the matrix W_{kk} is non-singular. Since Θ is non-singular, it follows that $\text{rank}(W_{kk}) = \text{rank}(w_{jk}\Theta^\top + w_{ki}\Theta^\top) = \text{rank}(w_{ji})$. Given that $w_{ji} = \begin{bmatrix} \text{diag}(\tilde{p}_{ji, \theta}^x) & 0 \\ 0 & \tilde{\phi}_{ji} \end{bmatrix}$, we conclude that $\text{rank}(W_{kk}) = 3$ if and only if $\tilde{p}_{ji, \theta}^x \neq 0$, $\tilde{p}_{ji, \theta}^y \neq 0$ and $\tilde{\phi}_{ji} \neq 0$.

Theorem IV.2. Under Assumptions IV.2 and III.2. If the leader velocity $\dot{g}_i^*(t)$ is time-varying and continuous, then the tracking errors δ_i and δ_f of the single-integrator multi-agent systems converge to zero globally and exponentially fast under the control law (28).

Proof: Under Assumption IV.2, W_{kk} is non-singular. The matrix-vector form of (28) is

$$\begin{cases} \dot{\delta}_l = -k_l \delta_l, \\ \hat{M}_{ff} \text{diag}(\Theta^\top) \dot{\delta}_f = -\hat{M}_{ff} \text{diag}(\Theta^\top) k_f \delta_f. \end{cases} \quad (29)$$

Since $\text{diag}(\Theta^\top)$ is non-singular and G satisfies Assumption III.2, according to Theorem III.1, we know that $\hat{M}_{ff}$ is non-singular, then we have $\dot{\delta}_f = -k_f \delta_f$. Additionally, $\dot{\delta}_l = -k_l \delta_l$, which implies that δ_l and δ_f globally converge to zero at an exponential rate. ■

Remark IV.2. Similar to Assumption 2 in [26], our Assumption IV.2 ensures the non-singularity of the matrix W_{kk} . While [18] requires the Laplacian to be positive semi-definite and satisfy a rank condition for this property, our approach instead imposes only a rank condition on the Laplacian, making the assumption substantially weaker.

B. Design of the Diagonal Stabilizing Matrix D

In the preceding section, we have obtained the global convergence of the proposed controller based on the existence of D . However, computing D remains a challenging inverse eigenvalue problem, which can be formulated as

$$\begin{aligned} &\text{find} && D = \text{diag}(x) \\ &\text{subject to} && \forall \lambda \in \sigma(D\hat{M}_{ff}), \\ &&& \Re(\lambda) > 0. \end{aligned} \quad (30)$$

where $x \in \mathbb{R}^{3(n-m)}$, $\sigma(\cdot)$ denotes the matrix spectrum.

This problem is inherently nonlinear, non-convex, and high-dimensional. Solving it typically requires centralized computation [11], [17], [46]. In this paper, we decompose the 2-rooted graph into multiple DEPs, enabling the computation of the stabilizing matrix to be performed independently for each DEP. This approach significantly reduces computational complexity. Furthermore, we derive explicit closed-form expressions for the diagonal matrix D in a DEP with $\ell = 1$ or 2, and rigorously prove that $D\hat{M}_{ff}$ exhibits strictly positive eigenvalues.

Theorem IV.3. Under Assumptions III.2 and IV.1, for a DEP-induced graph $\mathcal{L}_\kappa$. Each DEP $G_{\mathcal{P}_h} = (V_{\mathcal{P}_h}, E_{\mathcal{P}_h})$, $h = 1, \dots, \kappa$, with $\ell_h \in \{1, 2\}$ inner vertices, satisfies:

- If $\ell_h = 1$, with $V_{\mathcal{P}_h} = \{i, j, k\}$, $E_{\mathcal{P}_h} = \{(i, k), (j, k)\}$, then $D^h = w_{ij}^\top$, and $D^h \hat{M}_{ff}^h$ has positive eigenvalues.
- If $\ell_h = 2$, with $V_{\mathcal{P}_h} = \{i, j, k, l\}$, $E_{\mathcal{P}_h} = \{(i, k), (k, l), (l, k), (j, l)\}$, then

$$D^h = \begin{bmatrix} \text{sgn}(w_{il})|w_{kl}w_{ij}w_{kj}| + w_{il} & 0 \\ 0 & w_{kl}w_{ij}w_{il} \end{bmatrix}$$

and $D^h \hat{M}_{ff}^h$ has positive eigenvalues.

Thus, there exists a diagonal matrix $D = \text{diag}\{D^h\}$ such that $D\hat{M}_{ff}$ has positive eigenvalues, where w_{ij} is a diagonal matrix defined in (13) and $\hat{M}_{ff}$, $\hat{M}_{ff}^h$ are defined in (50).

Proof: Case $\ell_h = 1$: The matrix $\hat{M}_f^h$ for this configuration is

$$\hat{M}_f^h = \begin{bmatrix} \hat{M}_{fl}^h & \hat{M}_{ff}^h \end{bmatrix} = \begin{bmatrix} w_{jk} & w_{ki} & | & w_{ij} \end{bmatrix}. \quad (31)$$

Under Assumption III.2 and IV.1, we obtain

$$D^h \hat{M}_{ff}^h = w_{ij}^\top w_{ij}, \quad (32)$$

which is a positive definite matrix. Thus, all eigenvalues of $D^h \hat{M}_{ff}^h$ are strictly positive.

Case $\ell_h = 2$: The matrix $\hat{M}_f^h$ for this configuration is:

$$\hat{M}_f^h = \begin{bmatrix} \hat{M}_{fl}^h & \hat{M}_{ff}^h \end{bmatrix} = \begin{bmatrix} w_{lk} & 0 & | & w_{il} & w_{ki} \\ 0 & w_{lk} & | & w_{jl} & w_{kj} \end{bmatrix}. \quad (33)$$

Apply the permutation $P = [e_1, e_4, e_2, e_5, e_3, e_6]$, where e_i are standard basis vectors, to get $M' = P^T(D^h \hat{M}_{ff}^h)P = \text{diag}(A_x, A_y, A_\phi)$, where:

$$d_q = \text{sgn}(\tilde{p}_{il, \theta}^q) |\tilde{p}_{kl, \theta}^q \tilde{p}_{ij, \theta}^q \tilde{p}_{kj, \theta}^q| + \tilde{p}_{il, \theta}^q, \quad (34)$$

$$A_q = \begin{bmatrix} d_q \tilde{p}_{il, \theta}^q & d_q \tilde{p}_{ki, \theta}^q \\ \tilde{p}_{kl, \theta}^q \tilde{p}_{ij, \theta}^q \tilde{p}_{jl, \theta}^q & \tilde{p}_{kl, \theta}^q \tilde{p}_{ij, \theta}^q \tilde{p}_{il, \theta}^q \tilde{p}_{kj, \theta}^q \end{bmatrix}, \quad (35)$$

where $q \in \{x, y, \phi\}$, and for $q = \phi$, $\tilde{p}_{ij, \theta}^q = \tilde{\phi}_{ij}$ (i.e., the θ subscript is omitted).

For A_ϕ , under Assumption III.2 and IV.1, the trace is:

$$\text{tr}(A_\phi) = |\tilde{\phi}_{kl} \tilde{\phi}_{ij} \tilde{\phi}_{kj}| |\tilde{\phi}_{il}| + \tilde{\phi}_{il}^2 + \tilde{\phi}_{kl} \tilde{\phi}_{ij} \tilde{\phi}_{il} \tilde{\phi}_{kj} > 0. \quad (36)$$

Since $\tilde{\phi}_{il} \tilde{\phi}_{kj} - \tilde{\phi}_{jl} \tilde{\phi}_{ki} = \tilde{\phi}_{il} \tilde{\phi}_{kj} - \tilde{\phi}_{jl} (\tilde{\phi}_{kj} + \tilde{\phi}_{jl} + \tilde{\phi}_{li}) = \tilde{\phi}_{ij} \tilde{\phi}_{kl}$ and the Assumption III.2 and IV.1 hold, the determinant is:

$$\begin{aligned} \det(A_\phi) &= (\text{sgn}(\tilde{\phi}_{il}) |\tilde{\phi}_{kl} \tilde{\phi}_{ij} \tilde{\phi}_{kj}| + \tilde{\phi}_{il}) \tilde{\phi}_{kl}^2 \tilde{\phi}_{ij}^2 \tilde{\phi}_{il} \\ &= (|\tilde{\phi}_{il}| |\tilde{\phi}_{kl} \tilde{\phi}_{ij} \tilde{\phi}_{kj}| + \tilde{\phi}_{il}^2) \tilde{\phi}_{kl}^2 \tilde{\phi}_{ij}^2 > 0. \end{aligned} \quad (37)$$

For A_x and A_y , the analysis is analogous. Thus, all eigenvalues of $D^h \hat{M}_{ff}^h$ are strictly positive. ■

Theorem IV.3 proposes an approach for designing the diagonal stabilizing matrix when the sensing graph is a DEP-induced graph and each DEP has at most 2 inner vertices. However, when a DEP contains more than two inner agents, the matrix structure becomes more complex and may not admit the closed-form expression D^h . In such general cases, computing the stabilizing matrix still requires formulating an inverse eigenvalue problem (30). Nonetheless, since the stabilization can be performed independently for each DEP, the problem remains tractable and allows for decentralized or parallel computation.

V. A SIMULATION EXAMPLE

This section gives simulations to illustrate our results. We consider a nominal formation lying in $\mathbb{R}^2$ with DEP-induced graph shown in Fig. 2. The formation consists of two leaders $g_l = [g_1^\top, g_2^\top]^\top$ and seven followers $g_f = [g_3^\top, g_4^\top, g_5^\top, g_6^\top, g_7^\top, g_8^\top, g_9^\top]^\top$. The nominal configuration is given as: $\tilde{g}_1 = [-4, -2, \frac{\pi}{8}]^\top$, $\tilde{g}_2 = [2, 4, \frac{\pi}{4}]^\top$, $\tilde{g}_3 = [0, 1, \frac{\pi}{16}]^\top$, $\tilde{g}_4 = [1, -1, \frac{\pi}{16}]^\top$, $\tilde{g}_5 = [-1, 0, 0]^\top$, $\tilde{g}_6 = [-2, -3, \frac{3\pi}{16}]^\top$, $\tilde{g}_7 = [-3, -4, \frac{\pi}{4}]^\top$, $\tilde{g}_8 = [3, 2, \frac{\pi}{8}]^\top$, $\tilde{g}_9 = [4, 3, \frac{3\pi}{16}]^\top$, $\theta = 0$. The matrix-valued Laplacian $M(G, \tilde{g}, \theta)$ corresponding to the nominal formation $(G, \tilde{g}, \theta)$ can be calculated by (18), and the diagonal stabilizing matrix D can be obtained based on Theorem IV.3. It is clear that this nominal formation satisfies Assumptions III.2, IV.1, and IV.2.

This simulation aims to validate a proposed control strategy for coordinated formation control of multiple agents navigating dense obstacles. The control goal is to enable leaders to track the predefined reference trajectory, defined by maneuver parameters in Table I with cubic spline interpolation for continuously differentiable trajectories, while followers maintain a desired geometric formation using controller (28).

The simulation results, depicted in Fig. 7, illustrate the dynamic evolution of the formation. The initial positions and yaw angles of the agents are randomly assigned. Upon activation, the multi-agent system achieves the first target formation within 5 seconds. At this stage, the line formation ($s_x = 0$) and attitude alignment ($s_\phi = 0$) are established. During 5-10 seconds, the position formation executes pure translation. Subsequently (10-15s), s_ϕ transitions from 0 to -1, inducing an attitude scaling transformation. During 15-20s, the team navigates the obstacles by scaling the position formation ($s_y : 2.5 \rightarrow 4.5$) while maintaining the pre-configured attitudes from the previous phase. This scale-based avoidance strategy results in a tightly coordinated interplay between attitude and position formations, a capability unattainable by either technique in isolation, allowing the formation to navigate through a trumpet-shaped obstacle.

The formation then undergoes sequential maneuvers as follows.

- 20–25s: The attitude formation realigns and executes a $\frac{\pi}{4}$ translational shift.
- 25–30s: The position formation performs simultaneous translations in both x and y directions.

t(s)	s_x	τ_x	s_y	τ_y	s_ϕ	τ_ϕ
0	5	-47	2.5	4	1	0
5	0	-30	2.5	0	0	0
10	0	-15	2.5	0	0	0
15	0	-10	2.5	0	-1	0
20	0	0	4.5	0	-1	0
25	0	5	4.5	0	0	$\frac{\pi}{4}$
30	0	15	4.5	10	0	$\frac{\pi}{4}$
35	4.5	35	4.5	10	1	$-\frac{\pi}{4}$
45	2	50	2	-5	1	$-\frac{\pi}{4}$

TABLE I: Key maneuver parameters

- 30–35s: The formation performs a non-uniform scaling transformation, resulting in an enlarged formation pattern.
- 35–45s: The position formation performs a translation and a uniform contraction while maintaining fixed yaw angles.

By adjusting only leaders' positions and attitudes, the proposed control strategy enables continuous translations and non-uniform scalings of the joint position-attitude formation. Notably, by accounting for the physical shape of the agents (rather than modeling them as point masses), the proposed control method allows the formation to navigate through narrow arrays of parallel or non-parallel obstacles, as illustrated in the figure. In contrast, most existing approaches, e.g., [26], [44], [47]–[49], require the team to make a detour, resulting in reduced efficiency and a lower likelihood of finding feasible paths in dense obstacle environments.

As evidenced by Fig. 8, the tracking errors converge asymptotically to zero, validating the effectiveness of the proposed control strategy. This is consistent with the results established in Theorem IV.2.

VI. CONCLUSION

We have proposed a novel distributed leader-follower formation maneuver control framework for multi-agent systems in the plane, enabling simultaneous non-uniform scaling and translation of position and attitude formations. A matrix-valued Laplacian has been developed to characterize the target configuration space, and the nominal formation was shown to achieve maximum maneuverability if and only if the underlying sensing graph is 2-rooted. Additionally, by decomposing the graph into multiple DEPs, a scalable approach for the stabilizing matrix design was proposed. Simulation results have validated the effectiveness of the control strategy, showing that closed-loop errors converge globally to zero and adaptive formation maneuvers are achieved in dense obstacle scenarios. Future work will focus on designing controllers that leverage more sophisticated attitude transformations and enhance resilience to agent or edge failures, all without relying on a global coordinate system, thereby bridging the gap between theoretical advancements and practical deployment in real-world multi-agent systems.

VII. APPENDIX

A. Proof of Lemma II.1

(Sufficiency) Suppose G contains a spanning DEP-induced graph $\mathcal{L}_\kappa$ constructed recursively as in Definition II.4. By

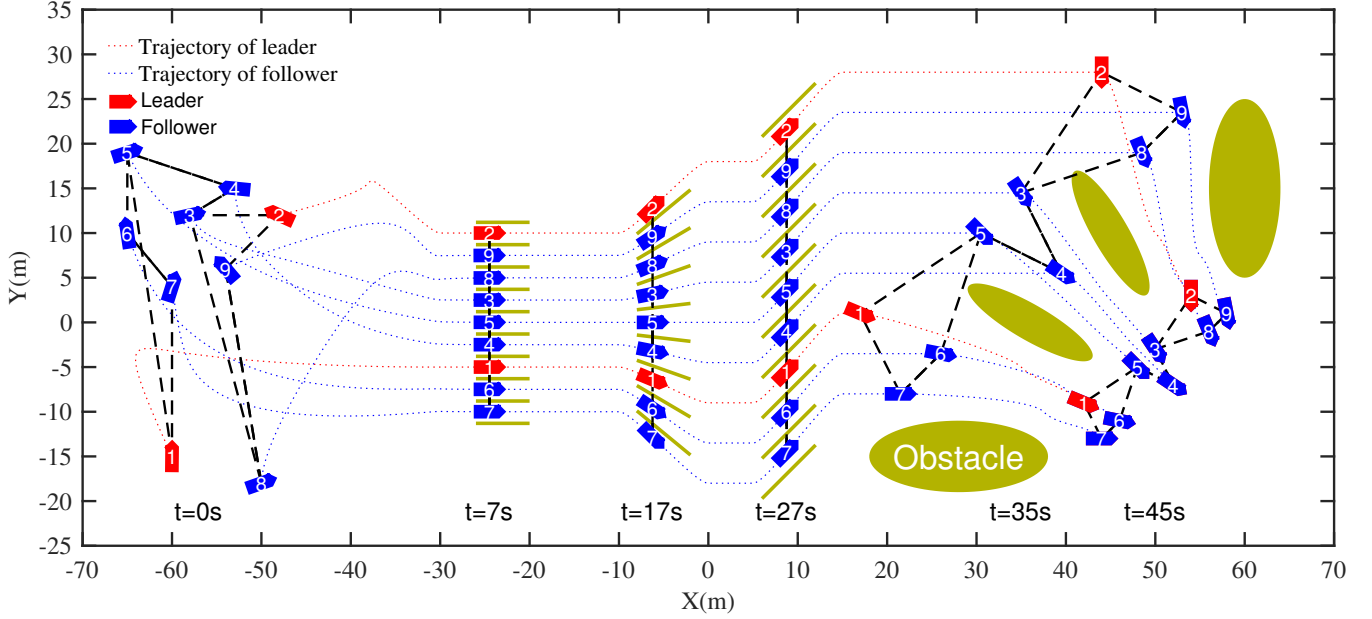


Fig. 7: Formation maneuver trajectories in 2-D space.

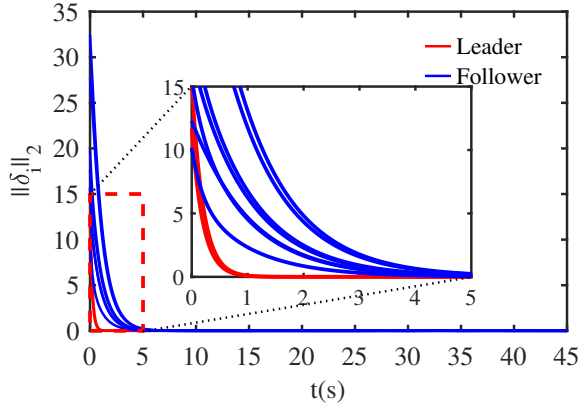


Fig. 8: Tracking errors.

definition, any agent in $\mathcal{G}_{\mathcal{P}_h}$ has two disjoint bidirectional paths in $\mathcal{L}_h$ to agents 1 and 2, respectively, $h = 1, \dots, \kappa$. Since $\mathcal{L}_\kappa$ spans G , every agent in G is 2-reachable from $\{1, 2\}$, i.e., G is 2-rooted.

(Necessity) Assume G is 2-rooted with roots $\{1, 2\}$. Initialize $\mathcal{L}_0 = (V_0, E_0)$, where $V_0 = \{1, 2\}$ and $E_0 = \emptyset$. For any agent $k \notin V_0$, since k is 2-reachable from $\{1, 2\}$, there must exist two disjoint paths from 1 and 2 to k , the union of these paths with involved vertices must contain a DEP $G_{\mathcal{P}_1} = (V_{\mathcal{P}_1}, E_{\mathcal{P}_1})$ with entry agents $i_1 = 1, j_1 = 2$ and $\ell_1 \geq 1$ inner agents labeled $\{|V_0|+1, \dots, |V_0|+\ell_1\}$. Construct $\mathcal{L}_1 = (V_1, E_1)$ with $V_1 = V_0 \cup V_{\mathcal{P}_1}$ and $E_1 = E_0 \cup E_{\mathcal{P}_1}$.

Next, select an agent $l \notin V_1$ that is 2-reachable from $\{1, 2\}$. There exist two disjoint paths from distinct agents $i_2, j_2 \in V_1$ to l , with all intermediate vertices distinct from V_1 . The union

of these paths with l must contain a DEP $G_{\mathcal{P}_2} = (V_{\mathcal{P}_2}, E_{\mathcal{P}_2})$ with entry agents $\{i_2, j_2\}$ and $\ell_2 \geq 1$ inner agents labeled $\{|V_1|+1, \dots, |V_1|+\ell_2\}$. Construct $\mathcal{L}_2 = (V_2, E_2)$ with $V_2 = V_1 \cup V_{\mathcal{P}_2}$ and $E_2 = E_1 \cup E_{\mathcal{P}_2}$.

Repeat the above process until all agents in V are included in some $\mathcal{L}_\kappa$. The resulting graph $\mathcal{L}_\kappa$ is a DEP-induced sub-graph by Definition II.4. ■

B. Proof of Lemma III.1

By Definition III.1, condition (a) holds if the mapping $[s^\top, \tau^\top]^\top \mapsto A[s^\top, \tau^\top]^\top = g$ is injective, i.e., $\text{null}(A) = \{0\}$. For $A \in \mathbb{R}^{3n \times 6}$, the rank-nullity theorem implies $\text{null}(A) = \{0\} \iff \text{rank}(A) = 6$ (condition (b)). Since $\Pi(\tilde{g}, \theta) = \text{image}(A)$, we have $\text{rank}(A) = 6 \iff \dim(\Pi(\tilde{g}, \theta)) = 6$ (condition (c)). Thus, conditions (a), (b), and (c) are equivalent. When $\text{rank}(A) < 6$, a singular configuration leads to ineffective parameters. Next, we prove the equivalence between condition (b) and condition (d) by establishing both implications.

Since $I_n \otimes \Theta$ is non-singular, we have

$$\text{rank}(A) = \text{rank}((I_n \otimes \Theta) \cdot \bar{A}) = \text{rank}(\bar{A}), \quad (38)$$

$$\text{where } \bar{A} = \begin{bmatrix} \text{diag}(\tilde{g}_{1,\theta}) & \Theta^\top \\ \text{diag}(\tilde{g}_{2,\theta}) & \Theta^\top \\ \vdots & \vdots \\ \text{diag}(\tilde{g}_{n,\theta}) & \Theta^\top \end{bmatrix} \in \mathbb{R}^{3n \times 6}.$$

(b) $\Rightarrow$ (d): Suppose $\{\tilde{p}_{i,\theta}^x\}_{i \in V}$ does not affinely span $\mathbb{R}$, i.e., $\tilde{p}_{i,\theta}^x = c$ for some constant c and all i . Then the first column of $\bar{A}$ is a constant vector:

$$v_1 = [c, 0, 0, c, 0, 0, \dots, c, 0, 0]^\top. \quad (39)$$

This vector can be written as a linear combination of the 4th and 5th columns of $\bar{A}$, denoted as v_4 and v_5 , respectively:

$$v_1 = -c \cos \theta v_4 + c \sin \theta v_5. \quad (40)$$

Hence, v_1 is linearly dependent on other columns, implying $\text{rank}(\bar{A}) = \text{rank}(A) < 6$. The same argument applies if $\{\tilde{p}_{i,\theta}^y\}_{i \in V}$ or $\{\tilde{\phi}_i\}_{i \in V}$ fails to affinely span $\mathbb{R}$. Therefore, all three sets must affinely span $\mathbb{R}$.

(d) $\Rightarrow$ (b): Assume each of the sets $\{\tilde{p}_{i,\theta}^x\}_{i \in V}$, $\{\tilde{p}_{i,\theta}^y\}_{i \in V}$, and $\{\tilde{\phi}_i\}_{i \in V}$ affinely spans $\mathbb{R}$. This implies the existence of distinct indices i_k, j_k ($k = 1, 2, 3$) such that:

$$\tilde{p}_{i_1,\theta}^x \neq \tilde{p}_{j_1,\theta}^x, \quad \tilde{p}_{i_2,\theta}^y \neq \tilde{p}_{j_2,\theta}^y, \quad \tilde{\phi}_{i_3} \neq \tilde{\phi}_{j_3}. \quad (41)$$

Next, apply rank-preserving operations to matrix $\bar{A}$: subtract row $3i_1 - 2$ from $3j_1 - 2$, row $3i_2 - 1$ from $3j_2 - 1$, and row $3i_3$ from $3j_3$. Consider the 6×6 submatrix with rows $3j_1 - 2$, $3j_2 - 1$, $3j_3$, $3i_1 - 2$, $3i_2 - 1$, $3i_3$:

$$\begin{bmatrix} \text{diag}([\tilde{p}_{j_1,\theta}^x - \tilde{p}_{i_1,\theta}^x, \tilde{p}_{j_2,\theta}^y - \tilde{p}_{i_2,\theta}^y, \tilde{\phi}_{j_3} - \tilde{\phi}_{i_3}]^\top) & 0 \\ \text{diag}([\tilde{p}_{i_1,\theta}^x, \tilde{p}_{i_2,\theta}^y, \tilde{\phi}_{i_3}]^\top) & \Theta^\top \end{bmatrix}. \quad (42)$$

Since $\tilde{p}_{j_1,\theta}^x - \tilde{p}_{i_1,\theta}^x, \tilde{p}_{j_2,\theta}^y - \tilde{p}_{i_2,\theta}^y, \tilde{\phi}_{j_3} - \tilde{\phi}_{i_3} \neq 0$, and $\Theta^\top$ is invertible, we have $\text{rank}(\bar{A}) = \text{rank}(A) = 6$. ■

C. Proof of Theorem III.1

The proof of Theorem III.1 requires a lemma.

Lemma VII.1. Consider a nominal formation $(G_{\mathcal{P}_h}, \tilde{g}, \theta)$ in $\mathbb{R}^2$, where $G_{\mathcal{P}_h} = (V_{\mathcal{P}_h}, E_{\mathcal{P}_h})$ is a DEP graph with entry agents $\{i_h, j_h\}$ and inner agents locally labeled $\{1, \dots, \ell_h\}$ as defined in Definition II.3. Then:

- 1) The matrix $\hat{M}_{ff}$ is non-singular if and only if the condition in (23) is satisfied.
- 2) Under conditions (23) and (26), there exists a diagonal matrix D such that every eigenvalue of $D\hat{M}_{ff}$ has a positive real part.

Proof: Let $g_\theta = \text{diag}(\Theta^\top)g = [\dots, g_{i,\theta}^\top, \dots]^\top$, where $g_{i,\theta} = \Theta^\top g_i = [p_{i,\theta}^x, p_{i,\theta}^y, \phi_i]^\top$. Define stacked vectors $p_\theta^x = [\dots, p_{i,\theta}^x, \dots]^\top$ and $p_\theta^y = [\dots, p_{i,\theta}^y, \dots]^\top$. Since each constant value matrix block w_{ij} defined in (13) is a diagonal matrix, there exist a row permutation matrix Q and a column permutation matrix P such that

$$\begin{aligned} Q\hat{M}_f P &= Q\hat{M}_f \begin{bmatrix} P_{ll} & 0 \\ 0 & P_{ff} \end{bmatrix} \\ &= \begin{bmatrix} \hat{M}_{fl}^x & 0 & 0 & \hat{M}_{ff}^x & 0 & 0 \\ 0 & \hat{M}_{fl}^y & 0 & 0 & \hat{M}_{ff}^y & 0 \\ 0 & 0 & \hat{M}_{fl}^\phi & 0 & 0 & \hat{M}_{ff}^\phi \end{bmatrix}. \end{aligned} \quad (43)$$

In other words, the matrix $\hat{M}_f$ can be decomposed into independent constraint matrices for each state component, we have

$$\begin{cases} \hat{M}_f^x p_\theta^x = 0, \\ \hat{M}_f^y p_\theta^y = 0, \\ \hat{M}_f^\phi \phi = 0, \end{cases} \quad (44)$$

where the matrices $\hat{M}_f^x = [\hat{M}_{fl}^x, \hat{M}_{ff}^x]$, $\hat{M}_f^y = [\hat{M}_{fl}^y, \hat{M}_{ff}^y]$, and $\hat{M}_f^\phi = [\hat{M}_{fl}^\phi, \hat{M}_{ff}^\phi]$ are partitioned according to the leader-follower structure. From (43), it holds that

$$Q\hat{M}_{ff}P_{ff} = \begin{bmatrix} \hat{M}_{ff}^x & 0 & 0 \\ 0 & \hat{M}_{ff}^y & 0 \\ 0 & 0 & \hat{M}_{ff}^\phi \end{bmatrix}. \quad (45)$$

Proof of Part 1): From (45), we have

$$\text{rank}(\hat{M}_{ff}) = \text{rank}(\hat{M}_{ff}^x) + \text{rank}(\hat{M}_{ff}^y) + \text{rank}(\hat{M}_{ff}^\phi). \quad (46)$$

Consequently, $\hat{M}_{ff}$ is non-singular if and only if $\hat{M}_{ff}^x$, $\hat{M}_{ff}^y$, and $\hat{M}_{ff}^\phi$ are all non-singular. Next, we establish the conditions under which $\hat{M}_{ff}^\phi$ is non-singular.

To simplify the notation, we adopt simplified indices by mapping the original agent labels $i_k, j_k, 1, 2, \dots, \ell_k$ to consecutive integers $1, 2, 3, \dots, n$. Under this notation, the matrix $\hat{M}_f^\phi$ takes the following form:

$$\hat{M}_f^\phi = \begin{bmatrix} \hat{M}_{fl}^\phi & \hat{M}_{ff}^\phi \end{bmatrix} = \begin{bmatrix} \phi_{43} & 0 & \phi_{14} & \phi_{31} & 0 & \dots & 0 \\ 0 & 0 & \phi_{54} & \phi_{35} & \phi_{43} & \ddots & \vdots \\ 0 & 0 & 0 & \phi_{65} & \phi_{46} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \phi_{n_1 n_2} \\ 0 & \phi_{nn_1} & 0 & \dots & 0 & \phi_{2n} & \phi_{n_1 2} \end{bmatrix}, \quad (47)$$

where $\hat{M}_{fl}^\phi \in \mathbb{R}^{(n-2) \times 2}$, $\hat{M}_{ff}^\phi \in \mathbb{R}^{(n-2) \times (n-2)}$, $\phi_{ij} = \phi_i - \phi_j$ and n_i is an abbreviation for $n - i$.

It is clear that $\hat{M}_{ff}^\phi = [m_{ij}]$ is a tridiagonal matrix. Let $f_0 = 1$, $f_1 = \det([\phi_{14}]) = \phi_{14}$, and $f_{n-2} = \det(\hat{M}_{ff}^\phi)$. According to [50, Theorem 2.1], $\det(\hat{M}_{ff}^\phi)$ can be computed from a three-term recurrence relation $f_i = m_{ii}f_{i-1} - m_{i(i-1)}m_{(i-1)i}f_{i-2}$, $i = 2, 3, \dots, n-2$ and f_i denotes leading principal minor of order i . Next, We prove this result by induction.

$$\begin{aligned} f_2 &= \phi_{35}f_1 - \phi_{54}\phi_{31}f_0 \\ &= (\phi_{34} + \phi_{45})f_1 - \phi_{54}\phi_{31} \\ &= \phi_{34}\phi_{14} + \phi_{45}(\phi_{14} + \phi_{31}) \\ &= \phi_{34}(\phi_{14} + \phi_{45}) \\ &= \phi_{34}\phi_{15}. \end{aligned} \quad (48)$$

Suppose $f_{n-4} = \phi_{34}\phi_{45} \dots \phi_{(n-3)(n-2)}\phi_{1(n-1)}$ and $f_{n-3} = \phi_{34}\phi_{45} \dots \phi_{(n-2)(n-1)}\phi_{1n}$, we have

$$\begin{aligned} f_{n-2} &= \phi_{(n-1)2}f_{n-3} - \phi_{2n}\phi_{(n-1)(n-2)}f_{n-4} \\ &= \phi_{(n-1)n}f_{n-3} + \phi_{n2}(f_{n-3} + \phi_{(n-1)(n-2)}f_{n-4}) \\ &= \phi_{(n-1)n}f_{n-3} + \phi_{n2}(\phi_{34}\phi_{45} \dots \phi_{(n-2)(n-1)}\phi_{1n} \\ &\quad + \phi_{34}\phi_{45} \dots \phi_{(n-3)(n-2)}\phi_{1(n-1)}\phi_{(n-1)(n-2)}) \\ &= \phi_{(n-1)n}f_{n-3} + \phi_{34}\phi_{45} \dots \phi_{(n-1)n}\phi_{n2} \\ &= \phi_{34}\phi_{45} \dots \phi_{(n-1)n}\phi_{12}. \end{aligned} \quad (49)$$

So, $\phi_{34}\phi_{45} \dots \phi_{(n-1)n}\phi_{12} \neq 0 \iff \det(\hat{M}_{ff}^\phi) \neq 0 \iff \text{rank}(\hat{M}_{ff}^\phi) = n - 2 \iff \hat{M}_{ff}^\phi$ is non-singular.

Similar to the above proof, we conclude that $\hat{M}_{ff}^x$ and $\hat{M}_{ff}^y$ are non-singular if and only if $\tilde{p}_{34,\theta}^x \tilde{p}_{45,\theta}^x \cdots \tilde{p}_{(n-1)n,\theta}^x \tilde{p}_{12,\theta}^x \neq 0$ and $\tilde{p}_{34,\theta}^y \tilde{p}_{45,\theta}^y \cdots \tilde{p}_{(n-1)n,\theta}^y \tilde{p}_{12,\theta}^y \neq 0$.

Proof of Part 2): Since $Q\hat{M}_{ff}P_{ff}$ is block-diagonal as shown in (43), we analyze the submatrices $\hat{M}_{ff}^x$, $\hat{M}_{ff}^y$, and $\hat{M}_{ff}^\phi$. By Lemma II.2, for each submatrix (e.g., $\hat{M}_{ff}^x$), there exists a diagonal matrix D^x such that every eigenvalue of $D^x\hat{M}_{ff}^x$ has a positive real part if its all leading principal minors are nonzero. The same applies to $\hat{M}_{ff}^y$ and $\hat{M}_{ff}^\phi$ with diagonal matrices D^y and D^ϕ , respectively. Construct $D = \text{diag}(D^x, D^y, D^\phi)$, which is diagonal and ensures that every eigenvalue of $DQ\hat{M}_{ff}P_{ff} = \text{diag}(D^x\hat{M}_{ff}^x, D^y\hat{M}_{ff}^y, D^\phi\hat{M}_{ff}^\phi)$ has a positive real part, since every eigenvalue of each block has a positive real part. Let $D' = DQ$, we note that since Q is a permutation matrix and D is diagonal, D' remains diagonal. Furthermore, $DQ\hat{M}_{ff}P_{ff}$ and $D'\hat{M}_{ff}$ share identical eigenvalues because P_{ff} is also a permutation matrix.

Next, we establish the conditions under which the leading principal minors of $\hat{M}_{ff}^x$, $\hat{M}_{ff}^y$, and $\hat{M}_{ff}^\phi$ are nonzero.

From the proof of Part 1), all leading principal minors of $\hat{M}_{ff}^\phi$ are distinct from zero $\iff f_1 \neq 0 \wedge f_2 \neq 0 \wedge \cdots \wedge f_{n-2} \neq 0 \iff \phi_{14} \neq 0 \wedge \phi_{34}\phi_{15} \neq 0 \wedge \cdots \wedge \phi_{34}\phi_{45} \cdots \phi_{(n-1)n}\phi_{12} \neq 0 \iff \phi_{34}\phi_{45} \cdots \phi_{(n-1)n}\phi_{14}\phi_{15} \cdots \phi_{1(n-1)}\phi_{1n}\phi_{12} \neq 0$. These conditions guarantee that $\hat{M}_{ff}^\phi$ has full rank and its leading principal minors are nonzero. The corresponding conditions for $\hat{M}_{ff}^x$ and $\hat{M}_{ff}^y$ follow similarly. ■

Proof of Theorem III.1: (Sufficiency) According to Definition II.4, the matrix $\hat{M}_f$ of DEP-induced graph $\mathcal{L}_\kappa$ takes the following form:

$$\hat{M}_f = [\hat{M}_{fl} \ \hat{M}_{ff}] = \begin{bmatrix} \hat{M}_{fl}^1 & \hat{M}_{ff}^1 & 0 & \cdots & 0 \\ * & * & \hat{M}_{ff}^2 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ * & * & \cdots & * & \hat{M}_{ff}^\kappa \end{bmatrix}, \quad (50)$$

where $\hat{M}_{fl} \in \mathbb{R}^{(3n-6) \times 6}$, $\hat{M}_{ff} \in \mathbb{R}^{(3n-6) \times (3n-6)}$, and $\hat{M}_{ff}^h$, $h \in \{1, 2, \dots, \kappa\}$ are the corresponding blocks of the DEP graph $\mathcal{G}_{\mathcal{P}_h}$. If G satisfies Assumption III.2, by applying Lemma VII.1, we have $\text{rank}(\hat{M}_{ff}^h) = 3|V_{\mathcal{P}_h}| - 6$. Considering the particular structure of $\hat{M}_f$, we know that

$$\text{rank}(\hat{M}_{ff}) = \sum_{h=1}^{\kappa} \text{rank}(\hat{M}_{ff}^h) = 3n - 6. \quad (51)$$

Given that $\hat{M}_{ff} \in \mathbb{R}^{(3n-6) \times (3n-6)}$ is a square matrix and its rank satisfies $\text{rank}(\hat{M}_{ff}) = 3n - 6$, it follows that $\hat{M}_{ff}$ is non-singular.

(Necessity) Suppose that G is not 2-rooted, implying that the removal of a particular agent results in some agents becoming unreachable from the root subset. For the sake of argument, assume that upon removing agent i , there emerges a subset U comprising $i - 1$ agents that are disconnected from all roots, and a complementary set $\bar{U}$ consisting of $n - i$ agents that remain accessible from at least one root. We can reindex the agents in U as $1, \dots, i - 1$ and those in $\bar{U}$ as $i + 1, \dots, n$.

Consequently, the matrix $\hat{M}_f^u$ associated with U adopts the following structure:

$$\begin{bmatrix} \hat{M}_{uu} & \hat{M}_{ui} & 0 \end{bmatrix}, \quad (52)$$

where $\hat{M}_{uu} \in \mathbb{R}^{(3i-3) \times (3i-3)}$ and $\hat{M}_{ui} \in \mathbb{R}^{(3i-3) \times 3}$. Denote the relabeled g by $[g_\alpha^\top, g_\beta^\top]^\top$ where $g_\alpha \in \mathbb{R}^{3i \times 1}$ and $g_\beta \in \mathbb{R}^{3(n-i) \times 1}$. By the definition of $\hat{M}_f$ and Lemma III.3, we have

$$[\hat{M}_{uu} \ \hat{M}_{uk}] \text{diag}(\Theta^\top) ((I_i \otimes S)g_\alpha + 1_i \otimes \tau) = 0. \quad (53)$$

This implies $\text{rank}([\hat{M}_{uu} \ \hat{M}_{ui}]) < 3i - 3$, meaning that $[\hat{M}_{uu} \ \hat{M}_{ui}]$ is not of full row rank. Consequently, $\hat{M}_f$ is not of full row rank, which entails that $\hat{M}_{ff}$ is singular. This contradicts the statement that $\hat{M}_{ff}$ is non-singular. Therefore, G is 2-rooted. According to Lemma II.1, G contains a spanning DEP-induced graph $\mathcal{L}_\kappa$. By (50) and Lemma VII.1, we conclude that if $\hat{M}_{ff}$ is non-singular, then the nominal formation $(G, \tilde{g}, \theta)$ must satisfy Assumption III.2. The proof is completed. ■

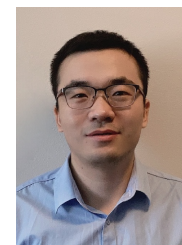
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